

MINERVA CryoCell P&ID v3 — Layer Naming Standard

Project: MINERVA CryoCell — SCK CEN (MYRRHA / MINERVA Phase 1)

Standards: ANSI/ISA-5.1-2022 · ISO 10628 · IEC 60617 · SCK CEN **AD_01.16**

Every v3 SVG uses the same ordered, hierarchical, self-describing Inkscape layer set. Layers are named `NN[_Sub]_Category_Detail` so they sort correctly and read unambiguously in Inkscape, Illustrator and any SVG editor. Drawing order is bottom (background) to top (annotations).

Layer table

#	Layer name	Purpose	Default colour	Line weight
00	00_Background_TitleBlock	Sheet frame, header strip, compact bottom title block	Black	0.7–2.0 mm
01	01_Scope_Boundaries	AD_01.16 termination-point (TP) scope diamonds	Per category	1.2 mm
02	02_Structure_Reference	Faded reference geometry from source	Grey	0.5 mm
03	03_Equipment_Vessels	Cavities, couplers, HX, vessels, tuners, antennae	Black / brown	0.5 mm
04A	04A_Piping_PRIMARY_40K	40 K shield trunk lines	Red #e00000	1.0 mm
04B	04B_Piping_BRANCHES_40K	40 K shield branch lines	Red #e00000	0.7 mm
05A	05A_Piping_PRIMARY_4p5K	4.5 K supply trunk lines	Blue #0033cc	1.0 mm
05B	05B_Piping_BRANCHES_4p5K	4.5 K supply branch lines	Blue #0033cc	0.7 mm
06A	06A_Piping_PRIMARY_2K	2 K return trunk lines	Cyan #00a6bd	1.0 mm
06B	06B_Piping_BRANCHES_2K	2 K return branch lines	Cyan #00a6bd	0.7 mm
07	07_Piping_SECONDARY_Water	DI cooling-water secondary circuit	Green #00a000	0.5 mm
08	08_Piping_OUTSIDE_SCOPE	60 K guard / infrastructure / instrument air	Grey dashed	0.35 mm
09	09_Valves_Mechanical		Black	1.0 mm

#	Layer name	Purpose	Default colour	Line weight
		CV/HV/SV/RV/ MV/PL valves + bellows		
10	10_Signals_Pneu matic	Pneumatic in- strument signals	Purple #7a00a0	0.25 mm
11	11_Signals_Elec tric	Electric instrument sig- nals	Blue #00529b	0.25 mm
12	12_Signals_Hydr aulic	Hydraulic instru- ment signals	Amber #a06a00	0.25 mm
13	13_Instruments_ Sensors	ISA instrument bubbles + heat- load markers	Per family	0.3 mm
14	14_Instruments_ Control_DIS	DIS interlock, tuner limit switches, Lemo connectors	Black / red	0.3–1.6 mm
15	15_Tags_Instrum ents	All tag text	Black	—
16	16_Legend_TOGGL EABLE	Compact legend overlay (off by default)	Mixed	—
17	17_Notes_TOGGLE ABLE	Buffer/scope notes and cal- louts	Green / red	—

Piping hierarchy — PRIMARY vs BRANCH

Within each cryogenic class (40 K, 4.5 K, 2 K) the source geometry is split into two sub-layers:

- **PRIMARY** (...A) — the trunk/header runs, drawn at **1.0 mm**.
- **BRANCH** (...B) — the take-offs and short connections, drawn at **0.7 mm**.

The classifier ranks every line segment by its on-sheet length (CTM-resolved). Segments at or above the 55th percentile of their class become PRIMARY; the rest become BRANCH. Junction nodes and coloured fills travel with the PRIMARY layer. Secondary (DI water) is a single 0.5 mm layer; out-of-scope services are a single 0.35 mm grey **dashed** layer.

Signal-line differentiation (all 0.25 mm)

To stay legible on monochrome plots each signal type uses a distinct dash pattern, following the intent of AD_01.16 Sheet 1/9:

Layer	Type	Pattern
10	Pneumatic	long dash + // cross-tick hatch marks
11	Electric	fine dotted
12	Hydraulic	dash-dot

Note on AD_01.16 vs the brief. AD_01.16 represents electrical as dashed, pneumatic with cross-ticks and hydraulic with an “L” mark. The brief asked for pneumatic = dashed, electric = dotted, hydraulic = dash-dot. v3 keeps all three **visually distinct** and leans on the AD_01.16 cross-tick for pneumatic while using dotted for electric (clearest on A3). The mapping is recorded here and in the on-sheet legend so there is no ambiguity.

Monochrome (_MONO) variants

Each sheet is also issued as a pure black-and-white _MONO SVG/PDF:

- all piping is **black**, differentiated only by the weight hierarchy above;
- out-of-scope services stay **dashed**; signals keep their dash patterns;
- instrument bubbles are **white-filled with black outline**;
- arrowhead / junction markers are recoloured to black.

Scope categories (AD_01.16)

Termination-point diamonds carry three compartments: TP / <letter><number> / <next system>. Category letters:

B = Building · C = Civil · E = Electrical · G = Compressed gasses ·
H = HVAC · L = Liquid waste · S = Solid waste · W = Water